

**Beyond Import Dependence:
Capability Closure and the Measurement of Strategic Black
Boxes**

PIATRA . INSTITUTE

August 2026

Abstract

In June 2026 the sole maker of extreme ultraviolet lithography systems stated that it had never shipped one to China and that none of the 314 machines operating worldwide is located there (Reuters, 2026a). Trade-based dependence measures therefore score Chinese exposure to that machine as zero, and in general they improve when a border closes. We separate import dependence from black-box dependence. A node is a strategic black box for a region when it is critical, when access to it can be interrupted by others, and when the region cannot credibly specify, verify, modify, produce, qualify, scale, service or replace it at the required performance, quantity, cost and horizon. In a model of 8 canonical node types with generative ground truth, a capability index recovers the true ordering of expected loss exactly (rank correlation 1.00), while import share and a concentration-weighted trade index correlate at -0.31 and -0.29 . Embargoing one node drops its trade index from 0.94 to 0 while its expected loss rises by a factor of 1.61. Treating capability as a stock, an import share of 0.5 builds capability above an absorptive capacity of 0.83 (without local engineering linkage) and erodes it below; the fall after hollowing takes 14 years and the climb back to the starting level 20. Two regions with identical mean burden of 0.2375 and opposite topology reverse their loss ranking with the type of shock, which argues against any scalar dependence index. The results support measuring capability and do not support closing borders.

Introduction

Extreme ultraviolet lithography is supplied by one firm, whose 2025 annual report records approximately 5,100 suppliers and 48 such systems shipped in the year, against 279 deep ultraviolet systems, down from 374 the year before (ASML 2025 Annual Report, 2026). The optics come exclusively from one supplier: more than 40,000 parts in the High-NA projection optics, weighing around twelve tons, more than 25,000 parts in the illumination system, a collaborating network of more than 1,200 members, and roughly a year to manufacture a single mirror (Carl Zeiss SMT, 2026). China imports none of these machines, because it is forbidden to. In June 2026 the manufacturer stated that it had never shipped one there, nor any component or module specially designed for one (Reuters, 2026a).

The European Commission's External Vulnerability Index, built on 2022 trade data, scores external exposure from import shares, supplier concentration and competitive position (Connell Garcia and Ho, 2025). Any measure of that family returns zero for Chinese exposure to this machine: there is no import to concentrate, no supplier to be exposed to and no flow to interrupt. The number is correct about the flow and uninformative about whether an industry can proceed while a capability remains out of reach.

The error has a direction. Trade measures are built from what crosses a border, so they improve when a border closes, while the quantity that governs consequences moves the other way.

Production and knowledge boundaries

Every production system has two boundaries that are commonly assumed to coincide. The production boundary concerns where a thing is made. The knowledge boundary concerns who could specify it, redesign it, verify that a substitute meets specification, manufacture it, qualify it for use, scale it, service it and build its successor. Competent system integrators routinely hold a knowledge boundary wider than their production boundary: firms know more than they make, deliberately, because integration requires understanding parts they have chosen not to build (Brunsoni, Prencipe and Pavitt, 2001). What a firm knows and what it makes have separable effects on performance under technological change (Kapoor and Adner, 2012).

Crossing the two boundaries gives four cells. A closed capability is made and understood in the region. A white-box import is made elsewhere and understood in the region, which is ordinary specialization and the condition of most trade. An enclave assembly is made in the region and understood elsewhere, so the output is domestic while the architecture, tooling, process recipes, firmware authority and certification are not. Full black-box dependence is made elsewhere and understood nowhere in the region. Trade statistics observe one axis of this space. They cannot separate the white-box import, the healthy case, from the black box, the dangerous one, and they score the enclave as domestic production.

What a supplier relationship transmits depends on how codifiable the transaction is, how complex the specification is and how capable the supplier is; the governance typology of global value chains predicts which relationships transmit knowledge and which transmit only parts (Gereffi, Humphrey and Sturgeon, 2005). Modular production networks were built to separate production from innovation, and they succeeded (Sturgeon, 2002). Some of what is separated is architectural knowledge, which is invisible in a component inventory and decisive when the architecture changes (Henderson and Clark, 1990). Some is tacit, carried in routines and practices and not in documents (Polanyi, 1966; Nelson and Winter, 1982). Holding a design without the complementary production and process assets is a known route to losing the returns on an innovation and eventually the innovation itself (Teece, 1986).

Definition of a strategic black box

A node is a strategic black box for a region, at a given horizon, performance level, production quantity, cost ceiling and coalition boundary, when three conditions hold together: the node gates output that matters; continued access can be materially influenced by firms, states, licences or routes outside the region; and the region or its coalition cannot credibly deliver qualified output at that performance, quantity and cost within that horizon after the incumbent supply stops.

Each qualifier matters. The status is relational, since a node is a black box for a particular region. It depends on time, since capabilities and frontiers move; on performance, since a working prototype falls short of a production tool; on quantity, since making ten differs from making a million; and on the coalition, since a member state, a union and an alliance have different degrees of closure. It is also probabilistic, since neither capability nor access is known exactly, so an assessment should

produce a distribution. A claim that a region “can make” something, stated without horizon, volume or yield, is incomplete.

The definition also lists what must be coded separately and is most often merged in practice: manufacturing location, corporate ownership, intellectual property, engineering knowledge, tooling access, software and update authority, certification authority, and control over allocation in a crisis. A foreign-owned plant on domestic soil is domestic in location and employment and may be foreign in every other respect. The reverse also occurs, and it is the more favourable case.

Comparison of dependence indices

The first mechanism expresses the argument quantitatively. Eight canonical node types receive generative ground truth that no index observes: the share of output the node gates, the annual probability that qualified supply is interrupted, the probability that the region can reconstitute it, whether a qualified alternative already exists, and how long reconstitution takes. An explicit process generates the expected discounted output loss over a 10-year horizon: interruptions arrive, an existing alternative is switched in quickly, the region rebuilds slowly if it can, and otherwise the loss continues to the end of the horizon. Three indices are scored against this loss. Import share, and a trade index formed from import share times supplier concentration, observe only flows. The capability index multiplies criticality, access risk, the complement of closure, the complement of substitutability, and reconstitution time normalized to the horizon.

The capability index recovers the true ordering exactly, with a rank correlation of 1.00 (Figure 1). Import share and the trade index score -0.31 and -0.29 . The trade measures are anti-correlated with the truth, because the node types that dominate expected loss are those with small flows. The embargoed frontier tool ranks first in true loss and last of 8 on the trade index. The enclave assembly, made locally under foreign design authority, ranks third in true loss and seventh on trade. The ore a region knows how to process ranks sixth in true loss and third on trade, an overstatement in the other direction, which matters because acting on it diverts funds from the real chokepoints.

The embargo case isolates the effect, because it holds the node fixed and changes only its legal status. The same tool, while supplied, has a trade index of 0.94 and an expected loss of 3.97. Embargoed, its trade index falls to 0 and its expected loss rises to 6.39, a factor of 1.61. A dashboard built on trade would record a large improvement in the week the exposure worsened.

The limitation is equally clear. Restricted to the four transparent nodes (the documented commodity, the white-box import, the understood ore and the midstream material), the trade index reaches a rank correlation of 0.80. Where nothing is opaque, flows approximate capability well and the additional machinery adds little. The capability index is needed in the opaque, controlled and enclave cases, which are the cases that motivate policy.

Types of black box

The definition covers cases with different remedies, which must be distinguished. A **resource chokepoint** lacks physical access to an ore, a chemical or an energy input, and its remedies are

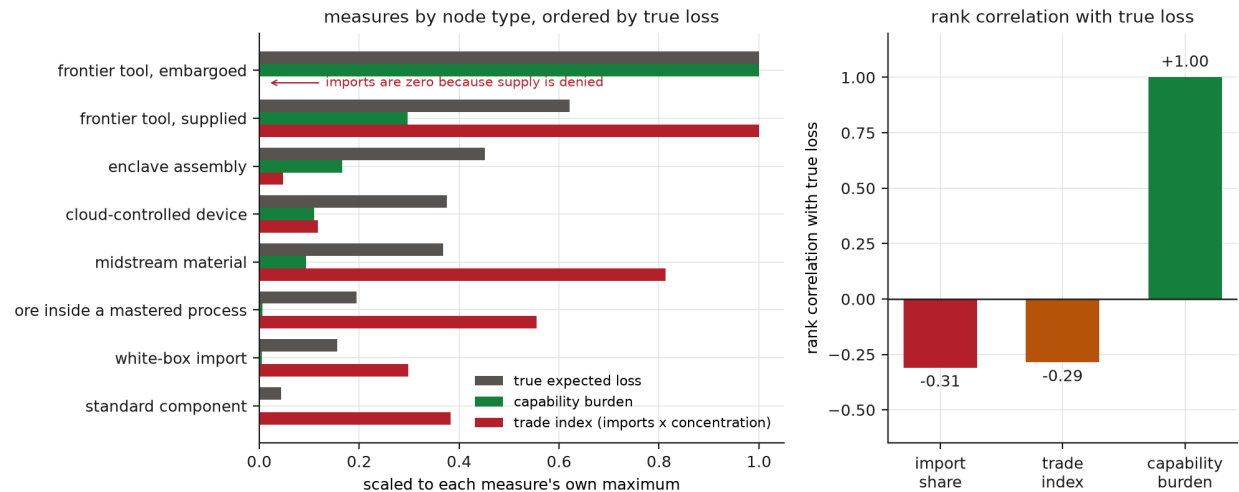


Figure 1: Comparison of dependence indices. Left: 8 canonical node types ordered by true expected loss, with each measure scaled to its own maximum. Right: rank correlation with true loss; the capability index recovers the ordering exactly while the two trade measures are negatively correlated with it.

diversification, recycling, inventory and substitution. A **process black box** lacks recipes, yield knowledge and quality control, and is addressed by pilot lines, production research and people who have run the process. A **tooling black box** lacks the machines and instruments needed to make the product, a recursive case since tools are made by tools. An **architectural black box** lacks the ability to specify interfaces and integrate a system, and is the hardest to notice, since every component may be available. A **software and control black box** lacks source code, update authority, licences or the cloud service a device depends on, and can be large while the associated goods trade is small. A **scale and qualification black box** has a prototype without yield, volume, uptime or certification.

Frontier lithography combines all six types except resource scarcity in one machine, which makes it the standard example and also misleading if read as the story of one machine. Senior Chinese semiconductor executives said as much in March 2026, describing progress in light sources, wafer stages and optical systems while integration into a working system remained unresolved, and naming design automation software, silicon wafers and electronic gases as bottlenecks alongside lithography (Reuters, 2026b). The controlled items confirm the breadth from the other side: the December 2024 United States rules cover 24 categories of semiconductor manufacturing equipment and 3 categories of development software, along with high-bandwidth memory (Bureau of Industry and Security, 2024). A list of items to deny is an estimate of the dependency graph, and it is much wider than one tool.

Dynamics of capability loss

Capability is a stock that depreciates. The second mechanism models it explicitly: closure grows with domestic production through learning by doing, with research, and with knowledge absorbed

from imports; it falls through depreciation and through the displacement of the domestic volume that sustained suppliers, tool shops, test rigs and trained engineers. Absorption is central, since using external knowledge requires prior related knowledge, so the capacity to absorb is itself a product of past investment (Cohen and Levinthal, 1990).

The model allows imports to have either sign, and the quantity of interest is where the sign changes (Figure 2). Without local engineering linkage, at an import share of 0.5, the break-even absorptive capacity is 0.83: above it, importing half of consumption leaves a region more capable than closing the border entirely, and below it the same import share erodes capability. Linkage lowers the threshold. An importer with absorptive capacity 0.75 and engineering linkage 0.5 at the same share ends at 0.686, against 0.665 under autarky, so openness with absorption outperforms closure. A passive importer at the same share ends at 0.351, and a deep importer at a share of 0.8 with low absorption falls to 0.132.

A minimum efficient scale converts erosion into collapse. Below it, suppliers exit, vocational programmes close, engineers leave, qualification infrastructure disappears and depreciation rises, the industrial-commons mechanism (Pisano and Shih, 2012). The deep, low-absorption importer crosses that threshold in its first year. A guaranteed production floor at 0.40 of demand, the modelled form of a procurement commitment, keeps the same region above the scale threshold and ends it at 0.306 instead of 0.132; the floor does not make the region a leader, but it keeps the commons alive.

The dynamics are hysteretic. The fall from the starting level takes 14 years to come within 0.02 of its final value. The climb back to the starting level, behind a protected market and a large public programme, takes 20 years, a factor of 1.43. The level held by a region that never lost the capability while importing with absorption cannot be reached behind the wall at all, because the protection that funds the rebuild also excludes the knowledge the open region was absorbing throughout. A region that hollows out therefore cannot return by closing to the position it would otherwise have held.

The same model separates deployment from learning. Two regions install identical capacity from imports at a share of 0.85, one with high absorption and local engineering linkage, the other turnkey. After 40 years their capability differs by 0.41 on a scale with maximum 1. Installed capacity and installed capability are different quantities, and only the first appears in deployment statistics.

Topology of dependence and the frontier

Regions differ in how dependence is arranged as well as in how much they carry. The third mechanism builds two regions with an identical mean burden of 0.2375 and opposite topology: one with 18 of 24 nodes above a moderate threshold, linked through shared upstream clusters, and one nearly closed except for 4 frontier nodes at extreme burden. Under shocks to a shared upstream cluster, the broad region loses 0.72 and the deep region 0.15. Under a shock that finds the single worst node, the deep region loses 0.86 and the broad region 0.30. The ranking reverses with the type of shock.

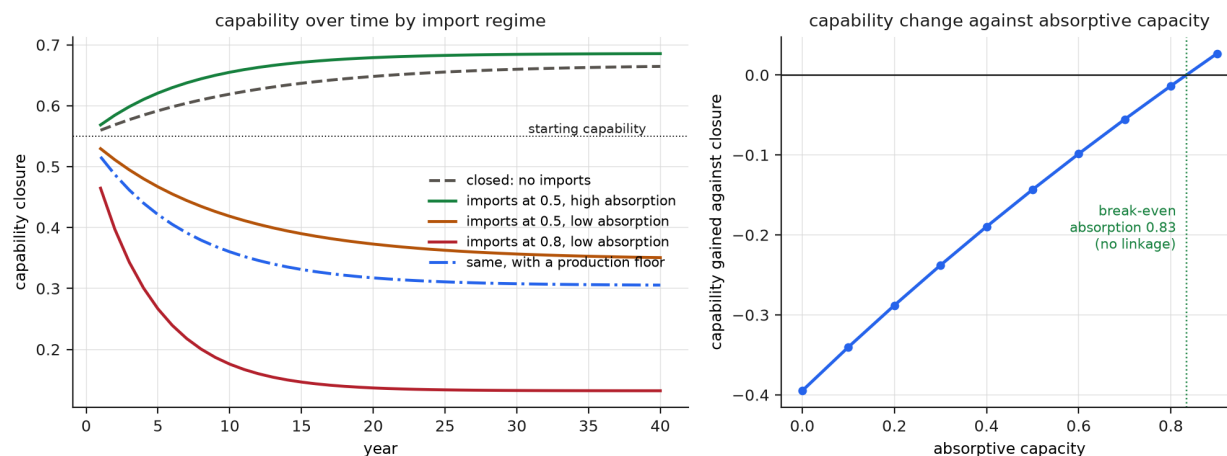


Figure 2: Dynamics of capability loss. Left: the same import share builds or erodes capability depending on absorptive capacity, with a production floor holding the commons above minimum efficient scale. Right: the sign of an import against absorptive capacity, with break-even at 0.83 without engineering linkage.

A single dependence number, however well constructed, cannot capture this. An average treats the two regions as identical, and a policy built on it is wrong for both. A region's assessment should report a distribution: how many nodes exceed a threshold, how severe the worst decile is, how concentrated the burden is, and how correlated the nodes are through shared upstream suppliers, since ten suppliers behind one cluster amount to one supplier. The empirical literature on production networks reaches the same conclusion: specific inputs propagate shocks, small nodes halt large outputs, and the aggregate consequences of microeconomic shocks depend on network structure more than on size (Acemoglu, Carvalho, Ozdaglar and Tahbaz-Salehi, 2012; Barrot and Sauvagnat, 2016; Carvalho, Nirei, Saito and Tahbaz-Salehi, 2021).

The moving frontier adds a further complication. A region can accumulate capability continuously while the target moves. In the modelled race, capability rises by a factor of 4.59 over 30 years and the relative position improves from 0.41 to 0.52 of the frontier, while the absolute gap widens from 0.62 to 1.79. Both statements are true, and public argument can be conducted entirely in terms of either. An assessment must therefore report the closure rate against the rate at which the frontier moves, and not capability alone. This describes the position of a lithography programme: the reported start of mass production of domestically developed immersion deep ultraviolet tools, at roughly 5 machines in 2026 and about 20 in 2027 with some critical components still imported, is real movement along the capability curve, while the frontier tool is a different machine (Reuters, 2026c).

Policy instruments

We compare four instruments per unit of spending against a mixed ensemble of shocks, since a planner does not know which type of shock will arrive. Diversification qualifies second sources; it is cheap per node and limited in effect, and it cannot help at a node whose burden is extreme partly

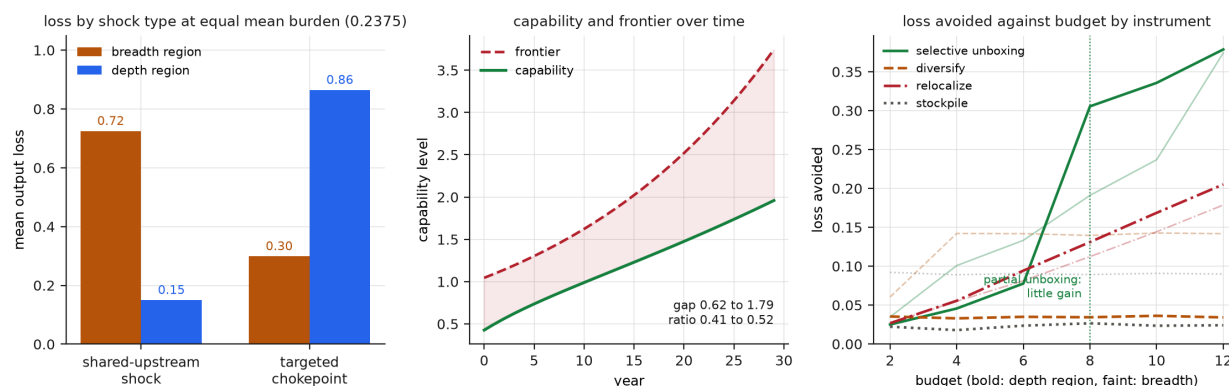


Figure 3: Topology and the frontier. Left: identical mean burden, opposite rankings under different shocks. Middle: capability improving as a ratio of the frontier while the absolute gap widens. Right: loss avoided per budget, bold for the deep region and faint for the broad one; rebuilding half of a set of chokepoints buys almost nothing.

because no alternative exists. Stockpiling is cheapest and buys months, so it does nothing against a node that needs a decade. Selective unboxing rebuilds the capability at a node, at high cost and nearly completely. Blanket relocalization spreads the same money across every node regardless of burden.

The results depend on topology. In the broad region, diversification performs best at moderate budgets, avoiding 0.14 of loss against 0.10 for unboxing at the same spending, because 18 moderate nodes can each receive a second source for the cost of 2 deep rebuilds. In the deep region, selective unboxing performs best only above a budget threshold. Below it, closing 2 of 4 chokepoints leaves the other 2, so partial rebuilding buys almost nothing, and in that underfunded range blanket relocalization scores best of the four because the alternatives are structurally unavailable. This result carries a qualification: the model charges all four instruments the same budget, while the modelled macroeconomic cost of relocalization, over 18 percent of global trade and over 5 percent of global real output, with volatility rising in more than half of the economies examined, is a cost the other three instruments do not carry (Organisation for Economic Co-operation and Development, 2025). With that cost included, the underfunded range argues for funding the chokepoints adequately.

The unit of policy is therefore a minimum viable capability portfolio: architecture, metrology, repair, second-source qualification, pilot-scale production, surge plans, open interfaces, and enough learning by doing to keep absorption high. A region can be very open and hold all of it. Tariffs restore none of it, because a tariff changes a price, and architectural knowledge, tacit process competence and certification authority are not priced goods.

Regional positions

The framework is used for comparison and implies no ranking of national merit. Europe holds nodes of great depth: the lithography integrator and its optics supplier are European capability commons with few equivalents elsewhere. Its characteristic risk is breadth, a larger number of

intermediate gaps across refined materials, midstream stages, components, tooling, platforms and qualification chains, which the topology mechanism shows to be exposed to correlated upstream shocks more than to single chokepoints. The concentration on the other side is documented: one country holds around 85 percent of solar and 80 percent of lithium-ion battery supply-chain capacity, 95 percent of photovoltaic wafers and 97 percent of anode materials, processes over 70 percent of lithium, cobalt, graphite and rare earths, and each of these chains contains at least one step where production outside the largest exporter covers less than a quarter of demand (International Energy Agency, 2026a). A month of interrupted battery-chain exports would cost about 17 billion United States dollars at electric-car plants elsewhere, almost two-thirds of it in the European Union.

Import share and missing capability are nevertheless different facts. Europe could make around 95 GW of inverters per year in 2025 against roughly 90 GW of demand, and switching utility-scale projects away from the dominant suppliers implies a project cost premium of just under 2 percent (International Energy Agency, 2026b). Conflating import share with absent capability produces expensive policy aimed at the wrong node. The trade index shows the same from the other direction: on 2022 data the European Union scores 0.22 on all industrial products against 0.28 for the United States and 0.13 for China, and 0.22 on semiconductors against 0.19 and 0.17 (Connell Garcia and Ho, 2025). On the flow measure Europe is not uniquely exposed. The claim here is that the flow measure is the wrong instrument.

China's characteristic risk is depth. Its stated programme addresses whole chains: the October 2025 Five-Year Plan recommendations call for decisive breakthroughs across entire chains in integrated circuits, machine tools, high-end equipment, basic software, advanced materials and biomanufacturing (Central Committee of the Communist Party of China, 2025). In the terms of this framework it is a plan to reduce black-box depth, and its unit of analysis is appropriate. It faces the frontier race described above, and depth is not confined to semiconductors: the C919 flies on an engine from a transatlantic joint venture (Safran, 2026), and in May 2025 a licence suspension turned that subsystem into a programme-level risk within weeks before being lifted in July. The dependency was invisible in ordinary statistics, became decisive by administrative decision, and reverted with the capability position unchanged, as the definition predicts.

The United States occupies a third position. Its share of global semiconductor manufacturing capacity fell from 37 percent in 1990 to 10 percent in 2022, and as of 2022 it produced no leading-edge logic at commercial scale and 3 percent of the two most common memory types (Government Accountability Office, 2025). In grid equipment, lead times reach two or more years and transformer prices rose by factors of 4 to 9 over 5 years for some categories (Department of Energy, 2025). This is a problem of reconstitution time, knowing how the equipment works while being unable to make enough of it soon enough, and not of epistemic opacity. It has different remedies, and an index that scored only closure would miss it, which is why reconstitution time enters the burden as a separate term.

For Latin America and Africa the framework uses capability ladders with four rungs: operating an installed system, maintaining it, reproducing it and redesigning it. Enclave sophistication means standing on the first rung while output statistics report the fourth, and the regional evidence de-

scribes persistent low growth, weak productivity and premature deindustrialization alongside genuine opportunities for upgrading (United Nations Industrial Development Organization, 2026). Pharmaceuticals illustrate the ladder: imports account for as much as 70 to 90 percent of medicines consumed in most of sub-Saharan Africa, and local production requires regulatory systems, analytical chemistry, validated process control and technology transfer in addition to plants (Dong and Mirza, 2016). The relevant question about an investment is which rung it moves a region to, the deployment-versus-learning result of the second mechanism applied to development.

Objections

The most serious objection is that capability scores are soft where trade data are hard. This is true, and it is why the definition is probabilistic and why coding rules matter: each dimension is scored separately, the evidence is recorded, contested nodes receive more than one assessor, reliability is reported, missing information is treated as uncertainty and not as zero, and announced capacity, laboratory prototypes and qualified serial production are kept in separate columns. A noisy measure of the right quantity can outperform a precise measure of the wrong one, and the first mechanism shows this: an index built from five coarse judgments recovers an ordering that flow data invert.

A second objection is that the mechanisms are stylized. The node types are constructed, the shock process is simple, and the burden formula is multiplicative because a node is dangerous only when several conditions hold together. The claims are accordingly about shape: which measure recovers an ordering, where the sign of an import changes, whether a ranking survives a change of shock, and whether an instrument's advantage depends on topology. Each survives changes in parameters, every constant is published, and every invariant is a test the run must pass.

A third objection is that the framework gives protectionism a vocabulary. The model's own results cut against this: closure loses to open absorption, blanket relocation is never the best instrument at an adequate budget, and the macroeconomic cost of relocation is stated alongside the result it qualifies. The claim is narrower: a region should know which of its nodes it could not reconstitute, and that knowledge is compatible with importing nearly everything.

A fourth objection is that access risk is political and cannot be forecast. This is why the term belongs in the index. Networks of interdependence become instruments of coercion for whoever controls their chokepoints (Farrell and Newman, 2019), so access probability depends on alignment and on other parties' decisions, and should be reported as a scenario over coalitions and not as a constant.

Falsification

The framework is empirical and fails under the following conditions. It fails if, once import concentration and network centrality are included, capability variables add no predictive power for output loss and recovery time after real shocks. It fails if domestic manufacturing presence predicts fast reconstruction regardless of ownership, tooling and architectural control, which would eliminate the enclave cell. It fails if external procurement produces no measurable long-term decay

of capability, which would remove the displacement term and the collapse of the commons. It fails if regional production volume is unrelated to supplier survival, skills and qualification infrastructure. It fails if capability scores cannot be reproduced across assessors, the one failure that would be fatal. And it fails if breadth and depth do not distinguish real regions, leaving the topology result without an empirical referent.

The study that would test these claims can be specified: an engineering ontology of nodes for each technology stack, in place of a list of trade codes; capability coded per dimension with recorded evidence and inter-rater reliability; and validation against dated disruptions in which output, price, lead time and recovery are observable, asking whether pre-shock black-box burden predicts severity better than import concentration and foreign value added. The present work supplies the framework and the mechanisms; the study has not been run.

Conclusion

Import statistics are exact about a quantity that ceases to be informative when a relationship becomes strategic, and they improve at the moment access is denied. The quantity that should be published is a region's credible ability to understand, verify, modify, reproduce, qualify, scale, service and supersede a node within the available time, together with the distribution of that ability across the region's nodes.

The hardest black box is the ecosystem that produces, measures, services and supersedes a product, which is why a country can hold every component of a machine and still be unable to build it, and why one firm's 5,100 suppliers and another's year-long mirror are the object of study more than the tool itself. The policy question is which of the things a region buys it could still make, and how long finding out would take.

Reproducibility

The simulation, its 3 figures and every model number quoted above regenerate from one command in the repository (`simulation/run_all.py`; mechanisms 1 and 3 run on recorded seeds and mechanism 2 is deterministic, bit for bit). 38 invariants fail the run if broken, among them the capability index outperforming both trade measures, the embargo simultaneously improving the trade index and worsening the true loss, the adequacy of the trade index on the transparent subset, the conditional sign of imports and its interior break-even, confirmed by bisection, the production floor preventing collapse of the commons, the rebuild taking longer than the decline while the never-lost level remains unreachable behind a wall, the ranking reversal between shock types, the widening frontier gap alongside an improving ratio, and the lumpiness of selective unboxing. Verified external figures enter `results.json` as cited records with their sources.

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